40 Principles with Explanation :

1. Segmentation (fragmentation) -

https://youtu.be/r7ZnJ4JC_IU?feature=shared
(TRIZ for Engineers - Page no. 154)
(Simplified TRIZ - Page no. 148)
Transition to microlevel. Divide an object or system into independent parts. Make an object easy to disassemble. Segmentation is a very frequently used principle. It helps to "combine the incompatible" and meet contradictory requirements in many different problems.

In fire extinguisher, more effective firefighting (good) requires more water and causes water damage (bad). Water is necessary and, at the same time, there should be no water. Let us segment water to small droplets and then increase the degree of fragmentation to mist. Mist can suppress fire effectively, using a very small amount of water. In previous chapters,

Examples:

a. Make cupcakes instead of one large cake so that people can decorate according to their own tastes and a variety of flavors can be offered.
b. Use powdered welding metal instead of foil or rod to get better penetration of a joint.
   https://www.youtube.com/watch?v=TqihlZG7HQo
c. Inject a drug in a finely powdered form.
d. Use Java applets. The predecessor of this example was the use of piping in UNIX to make large tasks possible by combining sequences of small tasks.
e. Paper is traditionally coated by a transferring application by a blade on a web.A new way is to spray coating in atomized form at high speed on both sides of the paper web.
   https://www.youtube.com/shorts/A8l4Txy_c8o

2. Extraction (Separation , taking out)

https://youtu.be/1HhKmQ0Hgos?feature=shared
(TRIZ for Engineers - Page no. 155)
(Simplified TRIZ - Page no. 150)
Separate the only necessary part (or property) or remove an interfering part or property or feature from an object or system

We need light, not lighting devices. Today, many parts of the lighting system are located some distance away from the places lighted. Reflectors and fiber optics both can separate the mechanism, such as a lamp, from the point of use of the light. Fiber optics are used in tiny surgical instruments to provide light exactly where it is needed inside the surgical area without the bulk of a lamp.

Examples :

a. Use a recording of a barking dog, without the dog, as a burglar alarm. https://www.youtube.com/shorts/Rfu8yp4JtaA
b. Use the sound of birds in distress instead of a scarecrow. https://www.youtube.com/shorts/NY6QzOlZdMs
c. We do not need a vacuum cleaner as such, but cleaning capacity. A central vacuum cleaning system leaves only nozzles and a piece of tubing in the apartment. Noisy and dirty parts are located where they do not disturb inhabitants.

3. Local quality

https://youtu.be/-ZX04F2zdaI?feature=shared
(TRIZ for Engineers - Page no. 156)
(Simplified TRIZ - Page no. 152)
Change an object's structure or an external environment (or external influence) so that the object will have different features or influences in different places or situations.

3.1. Change of an object's structure from uniform to non - uniform

Examples :

a. Drink cans shaped to facilitate stable stacking

b. Moulded hand grips on tools
c. Material surface treatments/coatings – plating, erosion/corrosion protection, case hardening,non - stick, etc
https://www.youtube.com/watch?v=xr53XIHaD64

3.2. Change an action or an external environment (or external influence) from uniform to non - uniform

Examples :

a. Introduce turbulent flow around an object to alter heat transfer properties
https://www.youtube.com/watch?v=ezcZNDR6M50

b. Use a gradient instead of constant temperature, density, or pressure
c. Take account of extremes of weather conditions when designing outdoor systems
d. Strobe lighting
https://www.youtube.com/watch?v=p4deMqX1CiI

3.3. Make each part of an object function in conditions most suitable for its operation

Examples :

a. Freezer compartment in refrigerator
b. Different zones in the combustion system of an engine
https://www.youtube.com/watch?v=5tN6eynMMNw
c. Night - time adjustment on a rear - view mirror
d. Lunch box with special compartments for hot and cold solid foods and for liquids

3.4. Make each part of an object fulfil a different and/or complementary useful function

Examples :

a. Swiss - Army knife
b. Combined can and bottle opener
c. Sharp and blunt end of a drawing pin
d. Rubber on the end of a pencil
https://www.youtube.com/shorts/BtPsQ5HMpws

e. Hammer with nail puller

4. Asymmetry (Symmetry change)

https://youtu.be/W8ZCdvw3vqM?feature=shared
(TRIZ for Engineers - Page no. 157)
(Simplified TRIZ - Page no. 153)

4.1. Change the shape or properties of an object from symmetrical to asymmetrical

Examples :

a. Asymmetrical funnel allows higher flow - rate than normal funnel
b. Put a flat spot on a cylindrical shaft to attach a locking feature
c. Oval and complex shaped O - rings
d. Coated glass or paper
e. Electric Plug
f. Introduction of angled or scarfed geometry features on component edges
g. Cutaway on a guitar improves access to high notes
h. Spout of a jug
   https://www.youtube.com/watch?v=xqmHhBh_wQc
i. Cam
j. Ratchet
k. Aerofoil – asymmetry generates lift.
l. Eccentric drive
   https://www.youtube.com/watch?v=WjR3AiiBHtw
m. Keys

4.2. Change the shape of an object to suit external asymmetries (e.g. ergonomic features)

Examples :

a. Human - shaped seating, etc
b. Design for left and right handed users
c. Finger and thumb grip features on objects
d. Spectacles
e. Car steering system compensates for camber in road
   https://www.youtube.com/watch?v=c6D46Zi69PE

f. Wing design compensated for asymmetric flow produced by propeller
https://www.youtube.com/watch?v=zwd9I_fIVZc
g. Turbomachinery design for boundary layer flows (‘ end - bend ’)
https://www.youtube.com/watch?v=SiOiVHUEYao

4.3. If an object is asymmetrical, increase its degree of asymmetry

Examples :

a. Use of variable control surfaces to alter lift properties of an aircraft wing
b. Special connectors with complex shape/pin configurations to ensure correct assembly
c. Introduction of several different measurement scales on a ruler

5. Combining (Merging , consolidation)

https://youtu.be/goAiMkqqui0?feature=shared
(TRIZ for Engineers - Page no. 158)
(Simplified TRIZ - Page no. 154)

Bring closer together (or merge) identical or similar objects; assemble identical or similar parts to perform parallel operations.

Examples :

a. Integration in microelectronics.
b. Telephone and computer networks.
c. Paper sheets constitute a book; books are merged into a library.
d. Fragile and weak components, such as glass plates, can be made stronger without increasing weight by combining them into packages.
e. An array of radio telescopes has greater resolving power than a single dish.
https://www.youtube.com/watch?v=fPJ2M7Caedg

6. Universality (Multifunctionality)

https://youtu.be/4vsCqTjCfHc?feature=shared
(TRIZ for Engineers - Page no. 159)
(Simplified TRIZ - Page no. 155)

Make a part of an object or system perform multiple functions, and eliminate the need for other parts . The number of parts and operations decreases and useful features and functions are retained.

Examples

a. Cross-functional training makes people much less susceptible to layoffs because they have multiple skills instead of one skill
b. One-stop shopping—supermarkets sell insurance, banking services, fuel, newspapers, etc.
c. Child's car safety seat converts to a pushchair https://youtu.be/XmVlF7QHCqo?feature=shared
d. Swiss Army knife https://www.youtube.com/shorts/N9QRpGmpVS8
e. Grill in a microwave oven
f. CD used as a storage medium for multiple data types
g. Cleaning strip at beginning of a cassette tape cleans tape heads

7. Nesting

https://youtu.be/VTi8kkZds7A?feature=shared
(TRIZ for Engineers - Page no. 160)
(Simplified TRIZ - Page no. 156)

Place one object inside another; place each object, in turn, inside the other. Make one part pass through a cavity in the other.

Note :

The name of this principle comes from the Russian folk art dolls or matrushkas, in which a series of wooden dolls are nested one inside the other

Examples :

a. Measuring cups or spoons https://www.youtube.com/watch?v=X26kAPrUG1E
b. The double hull in oil tankers https://www.youtube.com/watch?v=qgjVfdirGJY
c. Paint - brush attached to inside of lid of nail - varnish

8. Counterweight (Weight compensation)

https://youtu.be/QauNvSm1Gfw?feature=shared
(TRIZ for Engineers - Page no. 161)
(Simplified TRIZ - Page no. 157)

8.1. To compensate for the weight (downward tendency) of an object, merge it with other objects that provide lift

Example :

a. Kayak with foam floats built into hull cannot sink
b. Aerostatic aeroplane contains lighter - than - air pockets
c. Hot air or helium balloon.
   https://www.youtube.com/watch?v=P4awUtlWW94
d. Swim - bladder inside a fish
   https://www.youtube.com/watch?v=wxwfvU5-gt0
e. Flymo cutting blade produces lift

8.2. To compensate for the weight (downward tendency) of an object, make it interact with the environment (e.g. use aerodynamic, hydrodynamic, buoyancy and or global lift forces)

Examples :

a. Vortex generators improve lift of aircraft wings
   https://www.youtube.com/watch?v=-oUvi0dRlFU
b. Wing - in - ground effect aircraft
   https://www.youtube.com/watch?v=gRuuO1nelHQ
c. Hydrofoils lift ship out of the water to reduce drag
   https://www.youtube.com/watch?v=PXIy12snokE
d. Make use of centrifugal forces in rotating systems (e.g. Watt governor)
   https://youtu.be/zNzpOwWzxXw?feature=shared
e. Maglev train uses magnetic repulsion to reduce friction
   https://youtu.be/8os4A4li674?feature=shared

9. Prior counter-action (Preliminary counteraction)

https://youtu.be/N7S4e4CO37Y?feature=shared
(TRIZ for Engineers - Page no. 161)

(Simplified TRIZ - Page no. 158)

9.1. When it is necessary to perform an action with both harmful and useful effects, this should be replaced with anti - actions to control harmful effects

Examples :

a. Make clay pigeons out of ice or dung – they just melt away
b. Masking objects before harmful exposure: use a lead apron for X - rays
c. use masking tape when painting difficult edges etc.
d. Predict effects of signal distortion – compensate before transmitting
e. Buffer a solution to prevent harm from extremes of Ph

9.2. Create beforehand stresses in an object that will oppose known undesirable working stresses later on

Examples :

a. Pre - stress rebar before pouring concrete
b. Pre - stressed bolts
c. Decompression chamber to prevent divers getting the bends

10. Prior action (Preliminary action)

https://youtu.be/3svVa6410g8?feature=shared

(TRIZ for Engineers - Page no. 162)

(Simplified TRIZ - Page no. 159)

10.1. Perform the required change of an object in advance (either fully or partially)

Examples:

a. Precut parts for the building of wooden houses save work at the construction site.
b. Workers arrange their workspace so that the most frequently used tools (physical, paper, or electronic) are the easiest to reach.
c. Pre - pasted wall paper
d. Sterilize all instruments needed for a surgical procedure
e. Self - adhesive stamps https://www.youtube.com/watch?v=R_uQPYxh7nY
f. Holes cut before sheet - metal part formedPre - impregnated carbon fibre reduces lay - up time and improves “ wetting ”

g. Impregnated plasters
h. Chilled cocktail glasses

10.2 Pre - arrange objects such that they can come into action from the most convenient place and without losing time for their delivery

Examples:

a. Television demonstration chefs always have neat little dishes with all ingredients premeasured.
b. Manufacture flow - lines
c. Pre - deposited blad
d. Car jack, wheel brace, and spare tyre stored together
e. Collect all the tools and materials for the job before

11. Cushion in advance (Beforehand compensation)

https://youtu.be/MmOB88KZrSs?feature=shared
(TRIZ for Engineers - Page no. 163)
(Simplified TRIZ - Page no. 160)

Prepare emergency means beforehand to compensate for the relatively low reliability of an object or system over time

Note:

- Principles 9 through 11 together constitute a group of time-related principles, either for preventing problems or for correcting them quickly.
- Principle 11—commonly, users are told how to help themselves solve problems that are known to exist in the system.

Examples :

a. airbags in cars
b. pressure relief valves in boilers and chemical reactors.
c. Spare wheel/Battery back - up
d. Pressure relief valve
e. Emergency lighting circuit
f. Automatic save operations performed by computer programs
g. Crash barriers on motorways
h. first aid and rescue kits

i. fire extinguishers
https://www.youtube.com/watch?v=GVBamXXVD30

12. Equipotentiality (bring things to the same level)

https://youtu.be/_nx2yyNb36c?feature=shared
(TRIZ for Engineers - Page no. 164)
(Simplified TRIZ - Page no. 161)
Change operating conditions to eliminate the need to work against a potential field (e.g., eliminate the need to raise or lower objects in a gravity field).
Examples :

a. Spring loaded parts delivery system in a factory
b. Mechanic 's pit in a garage means car does not have to be lifted
c. Place a heavy object on ice, and let ice melt in order to lower it
d. Angle - poise lamp; changes in gravitational potential stored in springs
e. Descending cable cars balance the weight of ascending cars
f. Use grounding straps to bring people and objects to equal electrical potential to prevent harm from static electricity.

13. Inversion ("The other way around")

https://youtu.be/MJ-BcxJJHd8?feature=shared
(TRIZ for Engineers - Page no. 165)
(Simplified TRIZ - Page no. 162)
13.1. Invert the action used to solve the problem, (e.g. instead of cooling an object, heat it)
Examples :

a. To loosen stuck parts, cool the inner part instead of heating the outer part
b. Vacuum casting
c. Rotary engines
d. Test pressure vessel by varying pressure outside rather than inside
e. Test seal on a liquid container by filling with pressurised air and immersing in liquid; trails of bubbles are easier to trace than slow liquid leaks

f. Unstick frozen articles from each other by making them colder (rather than defrost)

13.2. Make movable parts (or the external environment) fixed, and fixed parts movable

Examples :

a. Hamster wheel
b. Rowing jogging machines
c. Rotate the part instead of the tool
d. Wind tunnels
e. Moving sidewalk with standing people
f. Drive through restaurant or bank

13.3. Turn the object (or process) ‘ upside down ’

Examples :

a. Tomato sauce bottle
b. Clean bottles by inverting and injecting water from below
c. Turn an assembly upside down to insert fasteners
d. Garage pit

14. Spheroidality (Curvature increase)

https://youtu.be/T5s5TcoUCQE?feature=shared

(TRIZ for Engineers - Page no. 166)

(Simplified TRIZ - Page no. 163)

14.1. Move from flat surfaces to spherical ones and from parts shaped as a cube (parallelepiped) to ball - shaped structures

Examples :

a. Use arches and domes for strength in architecture
b. Introduce fillet radii between surfaces at different angles
c. Introduce stress relieving holes at the ends of slots
d. Change curvature on lens to alter light deflection properties

14.2. Use rollers, balls, spirals

Examples :

a. Spiral gear (Nautilus) produces continuous resistance for weight lifting
b. Ball point and roller point pens for smooth ink distribution

c. Use spherical casters instead of cylindrical wheels to move furniture
d. Archimedes screw

14.3. Go from linear to rotary motion (or vice versa)
Examples :

a. Rotary actuators in hydraulic system
https://www.youtube.com/watch?v=F4j2wzbB8Uw

b. Switch from reciprocating to rotary pump
c. Push/pull versus rotary switches (e.g. lighting dimmer switch)
d. Linear motors
e. Linear versus rotating tracking arm on a record turntable ensures constant angle of stylus relative to groove
f. Screw - thread versus nail

14.4 . Use centrifugal forces
Examples :

a. Centrifugal casting for even wall thickness structures
b. Spin components after painting to remove excess paint
c. Remove water from clothes with a spin dryer rather than a mangle
d. Separate chemicals with different density properties using a centrifuge
e. Watt governor
f. Vortex/cyclone separates different density objects

15. Dynamicity (dynamicity, dynamization, dynamics)

https://youtu.be/t6AJ3vrj1K0?feature=shared
(TRIZ for Engineers - Page no. 167)
(Simplified TRIZ - Page no. 164)

15.1. Change the object (or outside environment) for optimal performance at every stage of operation
Examples :

a. Foam or Shape changing mattress
b. Gel fillings inside seat allow it to adapt to user
c. Adjustable steering wheel (or seat, or back support, or mirror position …)

d. Shape memory alloys/polymers
e. Racing car suspension adjustable for different tracks and driving techniques
f. Car handbrake adjustable to account for brake pad wear
g. Telescopic curtain rail – " one size fits all "
h. Easy to thread needle – eye opens for threading and closes again for use

15.2. Divide an object into parts capable of movement relative to each other

Examples :

a. Bifurcated bicycle saddle
b. Articulated lorry
c. Folding chair/mobile phone/laptop/etc
d. Collapsible structures
e. Brush seals

15.3. Change from immobile to mobile

Examples :

a. Bendy drinking straw
b. Flexible joint
c. Collapsible hose is flexible in use, and has additional flexibility of cross - section to make it easier to store

15.4. Increase the degree of free motion

Examples :

a. Use of different stiffness fibres in toothbrush – easily deflected at the edges to prevent gumdamage, hard in the middle
b. Flexible drive allows motion to be translated around bends
c. Loose sand inside truck tyre gives it self - balancing properties at speed
d. Add joints to robot arm to increase motion possibilities

16. Partial or overdone action (do a little less)

https://youtu.be/tN7QbcSVnZY?feature=shared

(TRIZ for Engineers - Page no. 168)

(Simplified TRIZ - Page no. 166)

If you can't achieve 100 percent of a desired effect – then go for more or less

Examples :

a. Over spray when painting, then remove excess

b. Fill, then ‘ top off ’ when pouring a pint of Guinness
c. Shrink wrapping process uses plastic deformation of wrapping to accommodate variations in
d. vacuum pressure
e. ‘ Roughing ’ and ‘ Finish ’ machining operations
f. Over - fill holes with plaster and then rub back to smooth
https://www.youtube.com/watch?v=Rk1bXo6mhAU

17. Moving to a new dimension (another dimension)

https://youtu.be/dKHxJw5mjFM?feature=shared
(TRIZ for Engineers - Page no. 169)
(Simplified TRIZ - Page no. 167)

17.1. Move into an additional dimension – from one to two – from two to three

Examples :

a. Coiled telephone wire
https://www.youtube.com/watch?v=4_CnQDbJURo

b. Curved bristles on a brush
c. Pizza - box with ribbed (as opposed to flat) base
d. Spiral staircase uses less floor area
https://www.youtube.com/watch?v=KF_fu1oUlvg

e. Introduction of down and up slopes between stations on railway reduces overall power requirements

17.2. Go single storey or layer to multi - storey or multi - layered

Examples :

a. Player with many CD ’ s
b. Stacked or multi - layered circuit boards
c. multi - storey Car parks & office blocks - Stacking
https://www.youtube.com/watch?v=notnDFsd3ow

17.3. Incline, tilt or re - orient the object, lay it on its side

Examples :

a. Cars on road transporter inclined to save space

17.4. Use ‘ another side ’ of a given area

Examples :

a. Press a groove onto both sides of a record
b. Mount electronic components on both sides of a circuit board
c. Print text around the rim of a coin
d. Paper clip – works by pressing both sides of paper together

18. Mechanical vibration

https://youtu.be/R8kKkmrtswY?feature=shared

(TRIZ for Engineers - Page no. 170)

(Simplified TRIZ - Page no. 168)

18.1. Cause an object to oscillate or vibrate

Examples :

a. Electric carving knife with vibrating blades
b. Shake/stir paint to mix before applying
c. Hammer drill
d. Vibration exciter removes voids from poured concrete
e. Vibrate during sieving operations to improve throughput
f. Musical instrument
g. Vibratory conveyor
h. Mobile phones

18.2. Increase its frequency (even up to the ultrasonic)

Examples :

a. Dog - whistle (transmit sound outside human range)
b. Ultrasonic cleaning
c. Non - destructive crack detection using ultrasound

18.3. Use an object ’s resonant frequency

Examples :

a. Destroy gallstones or kidney stones using ultrasonic resonance
b. Bottle cleaning by pulsing water jet at resonant frequency of bottles
c. Tuning fork
   https://www.youtube.com/watch?v=GQsOTf2DbKE

d. Increase action of a catalyst by vibrating it at its resonant frequency

18.4. Use piezoelectric vibrators instead of mechanical ones

Examples :

a. Quartz crystal oscillations drive high accuracy clocks
b. Piezoelectric vibrators improve fl uid atomisation from a spray nozzle
c. Optical phase modulator

18.5. Use combined ultrasonic and electromagnetic field oscillations (Use external elements to create oscillation/vibration)

Examples :

a. Mixing alloys in an induction furnace
b. Sono - chemistry
c. Ultrasonic drying of films – combine ultrasonic with heat source

19. Periodic action

https://youtu.be/Z9ADbTONXnw?feature=shared
(TRIZ for Engineers - Page no. 172)
(Simplified TRIZ - Page no. 169)

19.1. Instead of continuous action, use periodic or pulsating actions

Examples :

a. Hitting something repeatedly with a hammer
b. Pile drivers and hammer drills exert far more force for a given weight
c. Replace continuous siren with a pulsed sound
d. Pulsed bicycle lights make cyclist more noticeable to drivers
e. Pulsed vacuum cleaner suction improves collection performance
f. Pulsed water jet cutting
g. ABS car braking systems
   https://www.youtube.com/watch?v=hwwXukJaTlM

19.2. If an action is already periodic, change the periodic magnitude or frequency

Examples :

a. Improve a pulsed siren with changing amplitude and frequency
b. Dots and dashes in Morse Code transmissions
c. Use AM, FM, PWM to transmit information

19.3. Use pauses between actions to perform a different action

Examples :

a. Clean barrier filters by back - flushing them when not in use
b. Inkjet printer cleans heads between passes
c. Brush between suction pulses in vacuum cleaner
d. Multiple conversations on the same telephone transmission line
e. Use of energy storage means – e.g. batteries, fl y - wheels, etc

20. Continuity of a useful action

https://youtu.be/it5LR6V_Ihw?feature=shared

(TRIZ for Engineers - Page no. 173)

(Simplified TRIZ - Page no. 170)

20.1. Carry on work without a break. All parts of an object operating constantly at full capacity

Examples :

a. Flywheel stores energy when a vehicle stops, so the motor can keep running at optimum power
b. Constant output gas turbine in hybrid car, or APU in aircraft, runs at highest efficiency all the time it is switched on
c. Constant speed/variable pitch propeller
d. Self - tuning engine – constantly tunes itself to ensure maximum efficiency
e. Heart pacemaker
f. Improve composting process by continuously turning material
g. Continuous glass or steel production

20.2. Eliminate all idle or intermittent motion

Examples :

a. Self - cleaning/self - emptying filter eliminates down - time
b. Print during the return of a printer carriage – dot matrix printer, daisy wheel printers, inkjet printers
c. Digital storage media allow ‘ instant ’ information access
d. Kayaks use double - ended paddle to utilise recovery stroke
e. Computer operating systems utilise idle periods to perform necessary housekeeping tasks

21. Rushing through (Hurrying , skipping through)

https://youtu.be/6w2JGuzET3M?feature=shared
(TRIZ for Engineers - Page no. 174)
(Simplified TRIZ - Page no. 170)
Conduct a process, or certain stages of it (e.g. destructible, harmful or hazardous operations) at high speed
Examples :

a. Use a high speed dentist ' s drill to avoid heating tissue
b. Cut plastic faster than heat can propagate in the material, to avoid deforming the shape
c. Shatter toffee with a hammer blow
d. Drop forge
e. Flash photography
f. Super - critical shaft – run through resonant modes quickly

22. Convert harm into benefit ("Blessing in disguise")

https://youtu.be/QdtA5Wik2HA?feature=shared
(TRIZ for Engineers - Page no. 175)
(Simplified TRIZ - Page no. 172)
22.1. Harm into benefit – Use harmful factors (particularly, harmful effects of the environment or surroundings) to achieve a positive effect
Examples :

a. Use waste heat to generate electric power
b. Recycle scrap material as raw materials for another – e.g. chipboard
c. Vaccination
d. Lower body temperature to slow metabolism during operations
e. Composting
f. Use centrifugal energy in rotating shaft to do something useful – e.g. seal, or modulate cooling air
g. Use pressure differences to help rather than hinder seal performance

22.2. Eliminate the primary harmful action by adding it to another harmful action to resolve the problem

Examples :

a. Add a buffering material to a corrosive solution (e.g. an alkali to an acid, or vice versa)
b. Use a helium - oxygen mix for diving, to eliminate both nitrogen narcosis and oxygen poisoning from air and other nitrox mixes
c. Use gamma rays to detect positron emissions from explosives

22.3. Amplify a harmful factor to such a degree that it is no longer harmful

Examples :

a. Use a backfire to eliminate the fuel from a forest fire
b. Use explosives to blow out an oil - well fire
c. Laser - knife cauterises skin/blood vessels as it cuts

23. Feedback

https://youtu.be/BQBl2tEdbM0?feature=shared

(TRIZ for Engineers - Page no. 176)

(Simplified TRIZ - Page no. 174)

23.1. Introduce feedback to improve a process or action

Examples :

a. Automatic volume control in audio circuits
b. Signal from gyrocompass is used to control simple aircraft autopilots
c. Engine management system based on exhaust gas levels is more efficient than carburettor
d. Thermostat controls temperature accurately
e. Statistical Process Control – measurements are used to decide when to modify a process
f. Feedback turns inaccurate op - amp into useable accurate amplifier

23.2. If feedback is already used, change its magnitude or influence in accordance with operating conditions

Examples :

a. Change sensitivity of an autopilot when within 5 miles of an airport

b. Change sensitivity of a thermostat when cooling vs. heating, since it uses energy less efficiently when cooling
c. Use proportional, integral and/or differential control algorithm combinations

24. Mediator (Intermediary)

https://youtu.be/qI1fzfbS9Ik?feature=shared
(TRIZ for Engineers - Page no. 176)
(Simplified TRIZ - Page no. 175)
24.1. Use an intermediary carrier article or intermediary process
Examples :

a. Play a guitar with a plectrum
b. Use a chisel to control rock breaking/sculpting process
c. Dwell period during a manufacture process operation

Eradicate toxic cane toads (who were introduced to Sydney, Australia in 1935 from Hawaii and have defi ed all attempts to catch them) Trap the toads using ultraviolet disco lights. The lights attract insects which attract the toads)
24.2. Merge one object temporarily with another (which can be easily removed)
Examples :

a. Gloves to get hot dishes out of an oven
b. Joining papers with a paper clip
c. Introduction of catalysts into chemical reaction
d. Abrasive particles enhance water jet cutting
e. Bouquet garni in cooking

25. Self-service

https://youtu.be/Kk_farDsMdM?feature=shared
(TRIZ for Engineers - Page no. 177)
(Simplified TRIZ - Page no. 176)
25.1. An object must service itself by performing auxiliary helpful functions Halogen lamps regenerate the filament– evaporated material is re - deposited
Examples :

a. Self - aligning/self - adjusting seal

b. Self - cleaning oven/glass/material
c. Self winding watch + Watch face self heating
d. Odour eater
e. Anti bacterial Tupperware
https://www.youtube.com/watch?v=bHmkdAt5SOo
f. Distributor
g. Lizard's tail
h. Software patches - Anti - virus
i. Atomic clock
j. Auto Headlamp levelling
k. Self Loading rifle
https://www.youtube.com/shorts/za1tnxkq1m4

l. Tyres under golf course
m. Dissolvable stitches
n. Regenerative track steering
o. Trees photosynthesis
https://www.youtube.com/watch?v=CuAkjzwJOw0
p. Parthenogenesis
q. Self pollinating plants
r. Thru colour panels on Saturn cars
s. Penguins feet (stay frozen)
https://www.youtube.com/watch?v=zE2YNE376O8

t. Rock boots – wear away in a way that maintains grip
u. Customer data
v. Wikipedia
w. Self righting boats
x. Bone
y. FLAP valve
z. Dynomo on bike
aa. Regenerative braking
https://www.youtube.com/watch?v=stq2E3sZYg0
bb. Self closing door
https://www.youtube.com/watch?v=jl8BOsqcr1E

cc. Condensing boiler
https://www.youtube.com/watch?v=Xnl26qh5RXc

25.2. Use waste resources, energy, or substances
Examples :

a. Use heat from a process to generate electricity: Co - generation
b. Use animal waste as fertilizer
c. Use food and lawn waste to create compost
https://www.youtube.com/watch?v=9S9NpbZo6AE

d. Use pressure difference to reinforce seal action

26. Copying

https://youtu.be/i50WS4D87JU?feature=shared
(TRIZ for Engineers - Page no. 178)
(Simplified TRIZ - Page no. 177)
26.1. Replace unavailable, expensive, fragile object with inexpensive copies
Examples :

a. Imitation jewellery
b. Astroturf
c. Crash test dummy
d. Transfers instead of tattoos

26.2. Replace an object, or process with optical copies
Examples :

a. Do surveying from space photographs instead of on the ground
b. Measure an object by scaling measurements from a photograph
c. Virtual reality/Virtual mock - ups electronic pre- assembly modelling

26.3. If visible optical copies are used, move to infrared or ultraviolet copies
Examples :

a. Make images in infrared to detect heat sources, such as diseases in crops, or intruders in a security system

b. Use UV as a non - destructive crack detection method
c. UV light used to attract flying insects into trap

27. Inexpensive, short-lived object for expensive, durable one

https://youtu.be/Qowh0wyt50Y?feature=shared
(TRIZ for Engineers - Page no. 178)
(Simplified TRIZ - Page no. 178)
Replace an expensive object with a multiple of inexpensive objects, compromising certain qualities (such as service life, for instance)
Examples :

a. Cheaper tools – screwdrivers
b. Disposable serviettes, nappies paper cups/plates/forks/cameras/ torches /etc
c. Throw - away cigarette lighters
d. Matches versus lighters
e. Sacrificial coatings/components

28. Replacement of a mechanical system (use of fields)

https://youtu.be/kZ6NxIdy2Wg?feature=shared
(TRIZ for Engineers - Page no. 179)
(Simplified TRIZ - Page no. 178)
28.1. Replace a mechanical system with a sensory one
Examples :

a. Reversing sensors on cars vs. bumpers
b. Natural Gas odorisation
c. Replace a physical barrier with an acoustic one (audible to animals)
d. Add a bad smell to natural gas to alert users to leaks
e. Finger - print/retina/etc scan instead of a key
f. Voice activated telephone dialling
g. Movement sensors replace light switches
h. Fire alarms: triggered by smoke/ heat
i. Retinal recognition for security
j. Voice recognition software instead of typing

k. Traffic lights instead of traffic policeman
l. Lights and bells rather than secure barriers at rail crossings
m. Baby alarms
n. Security Systems

28.2. Use electric, magnetic and electromagnetic fields to interact with the object

Examples :

a. To mix 2 powders, electrostatically charge one positive and the other negative
b. Electrostatic precipitators separate particles from airflow
c. Improve efficiency of paint - spraying by oppositely charging paint droplets and object to be painted
d. Pick up tools
e. Magnetic bearings
f. Field activated switches
g. Boot lock replaced by electrical coil
h. TV remote
i. Maglev over wheels/rails

28.3. Replace stationary fields with moving, unstructured fields with structured

Examples :

a. Early communications used omni - directional broadcasting. We now use antennas with very detailed structure of the pattern of radiation
b. Magnetic Resonance Imaging (MRI) scanner

28.4. Use fields in conjunction with field - activated (e.g. ferromagnetic) particles

Examples :

a. Heat a substance containing ferromagnetic material by using varying magnetic field. When the temperature exceeds the Curie point, the material becomes paramagnetic, and no longer absorbs heat
b. Ferro - magnetic catalysts
c. Magneto - rheological effect – uses ferromagnetic particles and variable magnetic field to alter the viscosity of a fluid
d. Ferro - fluids – e.g. Magnatec oil – stay attached to surfaces requiring lubrication
e. Ultrasonic sealing (medical/plastic bags)

29. Pneumatic or hydraulic construction

https://youtu.be/_Gs3TqnG5gA?feature=shared
(TRIZ for Engineers - Page no. 180)
(Simplified TRIZ - Page no. 181)
Use gas and liquid parts of an object instead of solid parts (e.g. inflatable, filled with liquids, air cushion, hydrostatic, hydro - reactive)
Examples :

a. Transition from mechanical to hydraulic or pneumatic drive
b. Inflatable furniture/mattress/etc.
c. Gel filled saddle adapts to user
d. Hollow section O - rings
e. Hovercraft
f. Gas bearings
g. Acoustic panels incorporating Helmholtz resonators

30. Flexible membranes or thin film

https://youtu.be/62s4hnlEwUg?feature=shared
(TRIZ for Engineers - Page no. 180)
(Simplified TRIZ - Page no. 181)
30.1. Use flexible shells and thin films instead of three - dimensional structures
Examples :

a. Use inflatable (thin film) structures
b. Taut - liner trucks
c. Tarpaulin car cover instead of garage
d. Store energy in stretchable bags – accumulators in a hydraulic system

30.2. Isolate the object from its external environment using flexible membranes
Examples :

a. Bubble - wrap
b. Bandages/plasters
c. Tea bag
d. Heat curtain instead of solid door
e. Segmented plastic vertical strips used in refrigeration and hospitals

31. Use of porous material

https://youtu.be/nlKtq58Yfgk?feature=shared
(TRIZ for Engineers - Page no. 181)
(Simplified TRIZ - Page no. 182)
31.1. Make an object porous or add porous elements (inserts, coatings, etc.)
Examples :

a. Drill holes in a structure to reduce the weight
b. Cavity wall insulation
c. Transpiration film cooled structures
d. Foam metals
   https://www.youtube.com/watch?v=WuLHiZOkIvY
e. Use sponge - like structures as fluid absorption media
   https://www.youtube.com/watch?v=Qpt3NS6iXZ4

31.2. If an object is already porous, use the pores to introduce a useful substance or function
Examples :

a. Use a porous metal mesh to wick excess solder away from a joint
b. Store hydrogen in the pores of a palladium sponge. (Fuel " tank " for the hydrogen car – much safer than storing hydrogen gas)
c. Desiccant in polystyrene packing materials
d. Medicated swabs/dressings
   https://www.youtube.com/watch?v=sT_RUynntjA

32. Changing the color (Optical property changes)

https://youtu.be/tnHpAfY25ic?feature=shared
(TRIZ for Engineers - Page no. 182)
(Simplified TRIZ - Page no. 183)
32.1. Change the colour of an object or its external environment
Examples :

a. Colour - changing paint or suncream
b. Temperature - sensitive dyes used on food product labels to indicate when desired serving temperature has been achieved

c. Electrochromic glass
d. Light - sensitive glasses
e. Camouflage
f. Employ interference fringes on surface structures to change colour (as in butterfly wings, etc

32.2. Change the transparency of an object or its external environment

Examples :

a. Photolithography used to change transparent material to a solid mask for semiconductor processing
b. Light - sensitive glass

32.3. In order to improve observability of things that are diffi cult to see, use coloured additives or luminescent elements

Examples :

a. Fluorescent additives used during UV spectroscopy
b. UV marker pens used to help identify stolen goods
c. Use opposing colours to increase visibility – e.g. butchers use green decoration to make the red in meat look redder

32.4. Change the emissivity properties of an object subject to radiant heating

Examples :

a. Use of black and white coloured panels to assist thermal management on space vehicles
b. Use of parabolic reflectors in solar panels to increase energy capture
c. Paint object with high emissivity paint in order to be able to measure its temperature with a calibrated thermal imager

33. Homogeneity

https://youtu.be/UA1PaOvc5BI?feature=shared

(TRIZ for Engineers - Page no. 183)

(Simplified TRIZ - Page no. 184)

Objects interacting with the main object should be of same material (or material with identical properties)

Examples :

a. Gin ice cubes – Make ice - cubes out of the same fluid as the drink they are intended to cool

b. Food containers made of edible substances – ice cream cone
c. Container made of the same material as its contents, to reduce chemical reactions
d. Friction welding requires no intermediary material between the two surfaces to be joined
e. Temporary plant pots made out of compostable material
f. Human blood transfusions/transplants, use of bio - compatible materials
g. Join wooden components using (wood) dowel joints

34. Rejecting and regenerating parts

https://youtu.be/xsXfIOHbrQI?feature=shared
(TRIZ for Engineers - Page no. 183)
(Simplified TRIZ - Page no. 185)

34.1. Objects (or part of them) disappear after completing their useful function or becoming useless (discard them by dissolving, evaporating, etc) or modify them directly during the process

Examples :

a. Dissolving capsule for vitamins or medicine
b. Bio - degradable containers, bags, etc.
c. Casting processes – lost - wax, sand, etc.
d. Female spiders eat the male after mating
e. During firing of a rocket, foam protection is used on some elements; this evaporates in space when shock - absorbance is no longer required

34.2. Restore consumable/used up parts of an object during operation

Examples :

a. Self - sharpening blades – knives/lawn - mowers etc.
b. Strimmer dispenses more wire automatically after a breakage
c. Self - tuning automobile engines
d. Propelling pencil
e. Automatic rifle

35.Transformation of the physical and chemical states of an object (Optical property changes)

https://youtu.be/J0DYHMgXAik?feature=shared
(TRIZ for Engineers - Page no. 184)
(Simplified TRIZ - Page no. 187)

35.1. Change the physical state (e.g. to a gas, liquid, or solid)

Examples :

a. Transport oxygen or nitrogen or petroleum gas as a liquid, instead of a gas, to reduce volume

35.2. Change the concentration or density

Examples :

a. Liquid soap
b. Abradable linings used for gas - turbine engine seals

35.3. Change the degree of flexibility

Examples :

a. Vulcanize rubber to change its flexibility and durability
b. Compliant brush seals rather than labyrinth or fixed geometry seals

35.4. Change the temperature or volume

Examples :

a. Raise the temperature above the Curie point to change a ferromagnetic substance to a paramagnetic substance
b. Cooking/baking/etc.

35.5. Change the pressure

Examples :

a. Pressure cooker cooks more quickly and without losing fl avours
b. Electron beam welding in a vacuum
c. Vacuum packing of perishable goods

35.6. Change other parameters

Examples :

a. Shape memory alloys/polymers
b. Use Curie point to alter magnetic properties
c. Thixotropic paints/gels/etc
d. Use high conductivity materials – e.g. carbon fibre

36. Phase transformation

https://www.youtube.com/watch?v=f4O3tCDQL4E
(TRIZ for Engineers - Page no. 185)
(Simplified TRIZ - Page no. 187)
36.1 Use phenomena occurring during phase transitions. (e.g. volume changes, loss of absorption of heat etc.)
Examples :

a. Latent heat effects in melting/boiling
b. Soak rocks in water, then freezing causes water to expand – thus opening fissures in rock, making it easier to break
c. Heat pumps use the heat of vaporization and heat of condensation of a closed thermodynamic cycle to do useful work
d. Volume expansion during water - to - steam transition
e. Superconductivity

37. Thermal expansion

https://youtu.be/z-A6pHuEfrs?feature=shared
(TRIZ for Engineers - Page no. 186)
(Simplified TRIZ - Page no. 188)
37.1. Use thermal expansion (or contraction) of materials
Examples :

a. Fit a tight joint together by cooling the inner part to contract, heating the outer part to expand, putting the joint together, and returning to equilibrium
b. Metal tie - bars used to straighten buckling walls on old buildings
c. Thermal switch/cut - out
d. Shape memory alloys/polymers
e. Shrink - wrapping

37.2. Use multiple materials with different coefficients of thermal expansion
Examples :

a. Bi - metallic strips used for thermostats, etc
b. Two - way shape memory alloys
c. Passive blade tip clearance control in gas turbine engines

d. Combine materials with positive and negative thermal expansion coefficients to obtain alloys with zero (or specifically tailored) expansion properties – e.g. cerro - tru alloy used in the mounting and location of fragile turbine blade components during manufacture operations

38. Use strong oxidizers (accelerated oxidation)

https://youtu.be/mP5bXtns08U?feature=shared
(TRIZ for Engineers - Page no. 187)
(Simplified TRIZ - Page no. 189)

38.1. Replace common air with oxygen - enriched air

Examples :

a. Scuba diving with Nitrox or other non - air mixtures for extended endurance
b. Place asthmatic patients in oxygen tent
c. Nitrous oxide injection to provide power boost in high performance engines

38.2. Replace enriched air with pure oxygen

Examples :

a. Cut at a higher temperature using an oxy - acetylene torch
b. Treat wounds in a high pressure oxygen environment to kill anaerobic bacteria and aid healing
c. Control oxidation reactions more effectively by reacting in pure oxygen

38.3. Expose air or oxygen to ionising radiation

Examples :

a. Irradiation of food to extend shelf life
b. Use ionised air to destroy bacteria and sterilise food

38.4. Use ionised oxygen

Examples :

a. Speed up chemical reactions by ionising the gas before use
b. Separate oxygen from a mixed gas by ionising the oxygen

38.5. Replace ozonised (or ionised) oxygen with ozone

Examples :

a. Oxidisation of metals in bleaching solutions to reduce cost relative to hydrogen peroxide
b. Use ozone to destroy micro - organisms and toxins in corn

c. Ozone dissolved in water removes organic contaminants from ship hulls

39. Inert environment

https://youtu.be/ywFg9wz2saQ?feature=shared
(TRIZ for Engineers - Page no. 187)
(Simplified TRIZ - Page no. 190)
39.1. Replace a normal environment with an inert one
Examples :

a. Prevent degradation of a hot metal filament by using an argon atmosphere
b. MIG/TIG welding
c. Electron beam welding conducted in a vacuum
d. Vacuum packaging
e. Foam to separate a fire from oxygen in air

39.2. Add neutral parts, or inert additives to an object
Examples :

a. Naval aviation fuel contains additives to alter flash point
b. Add fire retardant elements to titanium to reduce possibility of titanium fire
c. Add foam to absorb sound vibrations – e.g. hi - fi speakers

40.Composite materials

https://youtu.be/1x0bPTUG8es?feature=shared
(TRIZ for Engineers - Page no. 188)
(Simplified TRIZ - Page no. 191)
Change from uniform to layered/composite (multiple) structures
Examples :

a. Aircraft structures where low weight and high strength are required
b. Composites in golf club shaft
c. Concrete aggregate
d. Glass - reinforced plastic
e. Fibre - reinforced ceramics
f. Hard/soft/hard multi - layer coatings to improve erosion properties